

SURFACE TENSION

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CAN YOU RECALL?

- ◉ You must have blown soap bubbles in your childhood. What is their shape?
- ◉ Why does a needle float on the surface of water?
- ◉ Why can a water spider walk comfortably on the surface of still water?
- ◉ Why are free liquid drops and bubbles always spherical in shape?

The answer for all the above questions is
SURFACE TENSION!!!!

SURFACE TENSION

A liquid at rest shows a very interesting property called *surface tension*.

- ⊙ water spider walks on the surface of steady water
- ⊙ A needle floats on the steady surface of water
- ⊙ rain drops, soap bubbles and mercury always take Spherical shape
- ⊙ All these phenomena arise due to surface tension.
- ⊙ Surface tension is one of the important properties of liquids.



Surface tension of some liquids at 20°C.

Sr. No.	Liquid	S.T. (N/m)	S.T. (dyne/cm)
1	Water	0.0727	72.7
2	Mercury	0.4355	435.5
3	Soap solution	0.025	25
4	Glycerin	0.0632	63.2

DO YOU KNOW?

1. When we write on paper, the ink sticks to the paper.
 2. When teacher writes on a board, chalk particles stick to the board.
 3. Mercury in a glass container does not wet its surface, while water in a glass container wets it.
- All the above observations can be explained on the basis of different types of forces coming into play in all these situations.
 - We will try to understand the effect of these forces and their relation to the surface tension in liquids.

MOLECULAR THEORY OF SURFACE TENSION:

⊙ To understand surface tension, we need to know some terms in molecular theory that explain the behaviour of liquids at their surface.

a) Intermolecular force: Matter is made up of molecules. Any two molecules attract each other. *This force between molecules is called intermolecular force.*

There are two types of intermolecular forces –

- i) Cohesive force
- ii) Adhesive force.

COHESIVE FORCE

- ◉ The force of attraction between the molecules of the same substance is called cohesive force or force of cohesion.
- ◉ Occurs between similar molecules
- ◉ Includes Van Der Waal forces and hydrogen bonding
- ◉ Between water molecules



ADHESIVE FORCE

- ◉ The force of attraction between the molecules of different substances is called adhesive force or force of
- ◉ Occurs between different molecules
- ◉ Includes electrostatic attraction.
- ◉ Between water molecules and surface of leaf.

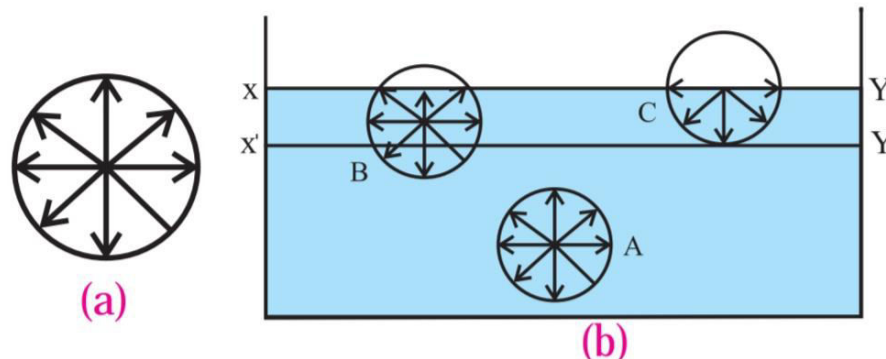


Range of molecular force: The maximum distance from a molecule up to which the molecular force is effective is called the range of molecular force.

Intermolecular forces are effective up to a distance of the order of **few nanometer (10^{-9} m)** in solids and liquids. Therefore, they are short range forces.

Sphere of influence: An imaginary sphere with a molecule at its center and radius equal to the molecular range is called the sphere of influence of the molecule. The spheres around molecules A, B or C are shown in Fig. (a) and (b).

The intermolecular force is effective only within the sphere of influence.

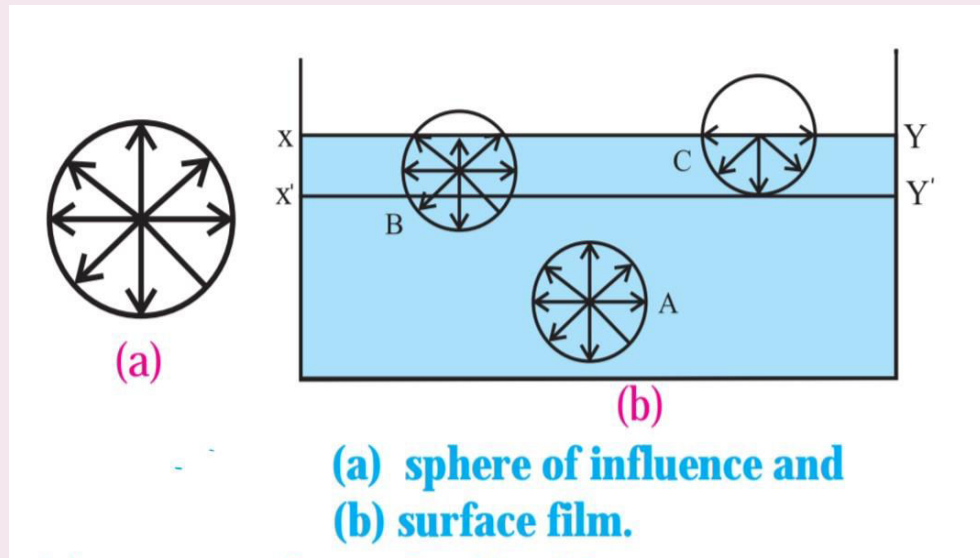


(a) sphere of influence and
(b) surface film.

◎ **Surface film:** The surface layer of a liquid with thickness equal to the range of intermolecular force is called the surface film. This is the layer shown between XY and $X'Y'$ in Fig. (b).

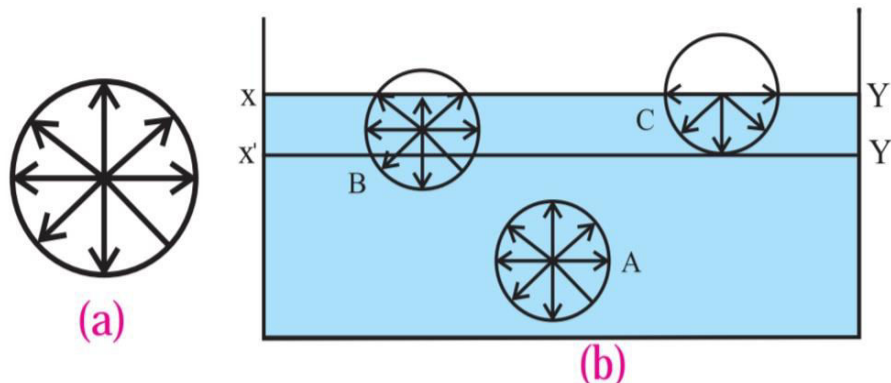
◎ **Free surface of a liquid:** It is the surface of a fluid which does not experience any shear stress.

For example, the interface between liquid water and the air above. In Fig. (b), XY is the free surface of the liquid.



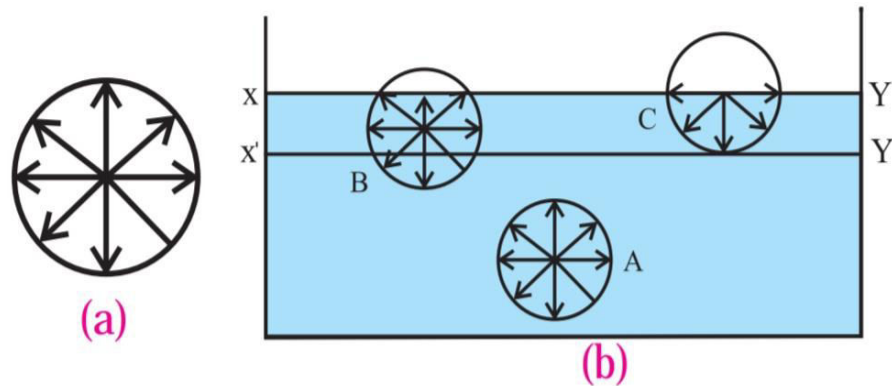
SURFACE TENSION ON THE BASIS OF MOLECULAR THEORY:

- As shown in Fig. (b), the section $XX'-Y'Y$ near the surface of the liquid acts as the surface film.
- Consider three molecules A, B, and C such that molecule A is deep inside the liquid, molecule B within surface film and molecule C on the surface of the liquid.
- As molecule A is deep inside the liquid, its sphere of influence is also completely inside the liquid. As a result, molecule A is acted upon by equal cohesive forces in all directions. *Thus, the net cohesive force acting on molecule A is zero.*



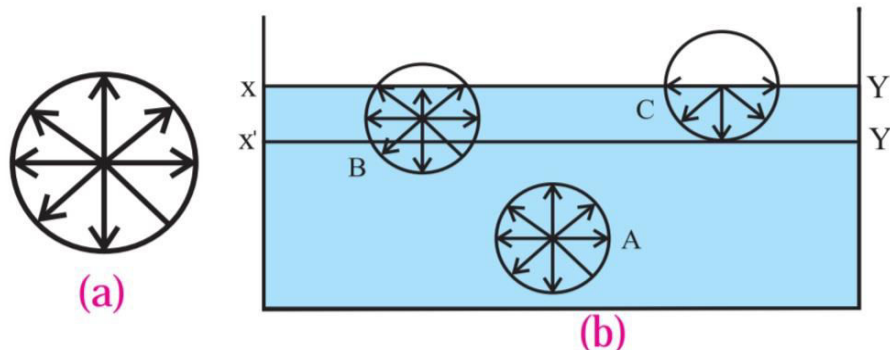
(a) sphere of influence and
(b) surface film.

- ⊙ Molecule B lies within the surface layer and below the free surface of the liquid.
- ⊙ A larger part of its sphere of influence is inside the liquid and a smaller part is in air. Due to this, a strong downward cohesive force acts on the liquid molecule.
- ⊙ The adhesive force acting on molecule B due to air molecules above it and within its sphere of influence is weak. It points upwards. As a result, the molecule B gets attracted inside the liquid.



**(a) sphere of influence and
(b) surface film.**

- ◉ The same holds for molecule C which lies exactly on the free surface of the liquid.
- ◉ Half of the sphere of influence is in air and half in the liquid. The number of air molecules within the sphere of influence of the molecule C, above the free surface of the liquid is much less than the number of liquid molecules within the sphere of influence that lies within the liquid.
- ◉ This is because, the density of air is less than that of a liquid. The adhesive force trying to pull the molecule above the liquid surface is much weaker than the cohesive force that tries to pull the molecule inside the liquid surface.
- ◉ *As a result, the molecule C also gets attracted inside the liquid.*



(a) sphere of influence and
(b) surface film.

- ⊙ Thus, all molecules in the surface film are acted upon by an unbalanced net cohesive force directed into the liquid.
- ⊙ Therefore, the molecules in the surface film are pulled inside the liquid.
- ⊙ This minimizes the total number of molecules in the surface film.
- ⊙ As a result, the surface film remains under tension.
- ⊙ The surface film of a liquid behaves like a stretched elastic membrane.
- ⊙ This tension is known as surface tension.
- ⊙ *The force due to surface tension acts tangential to the free surface of a liquid.*

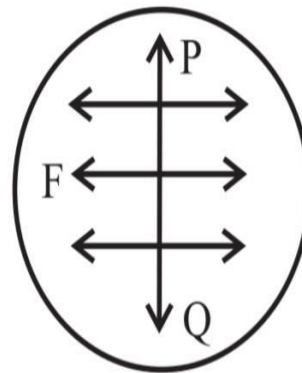
SURFACE TENSION AND SURFACE ENERGY:

⊙ Surface Tension:

The free surface of a liquid in a container acts as a stretched membrane and all molecules on the surface film experience a stretching force.

Imagine a line PQ of length L drawn tangential to the free surface of the liquid, as shown in Figure.

- ⊙ All the molecules on this line experience equal and opposite forces tangential due to the cohesive force.



Force of surface tension.

⊙ This force per unit length is the surface tension. *Surface tension T is defined as, the tangential force acting per unit length on both sides of an imaginary line drawn on the free surface of liquid.*

$$T = \frac{F}{L}$$

SI unit of surface tension is N/m. Its Dimension are, $[L^0M^1T^{-2}]$.

⊙ **SURFACE ENERGY:** We have seen that a molecule inside the volume of a liquid experiences more cohesive force than a molecule which are on the surface film of the liquid.

⊙ Thus, work has to be done to bring any molecule from inside the liquid into the surface film.

⊙ Clearly, the surface molecules possess extra potential energy as compared to the molecules inside the liquid.

⊙ The extra energy of the molecules in the surface layer is called the *surface energy* of the liquid.

⊙ As any system always tries to attain a state of minimum potential energy, the liquid tries to reduce the area of its surface film. Energy has to be spent in order to increase the surface area of a liquid.

REMEMBER THIS

- 1) Molecules on the liquid surface experience net inward pull. In spite of this if they remain at the surface, they possess higher potential energy. As a universal property, any system tries to minimize its potential energy. Hence liquid surface tries to minimize its surface area.
- 2) When a number of droplets coalesce and form a drop, there is reduction in the total surface area. In this case, energy is released to the surrounding.

RELATION BETWEEN THE SURFACE ENERGY AND SURFACE TENSION

- Consider a rectangular frame of wire $P'PSS'$. It is fitted with a movable arm QR as shown in Fig.
- This frame is dipped in a soap solution and then taken out. A film of soap solution will be formed within the boundaries $PQRS$ of the frame.

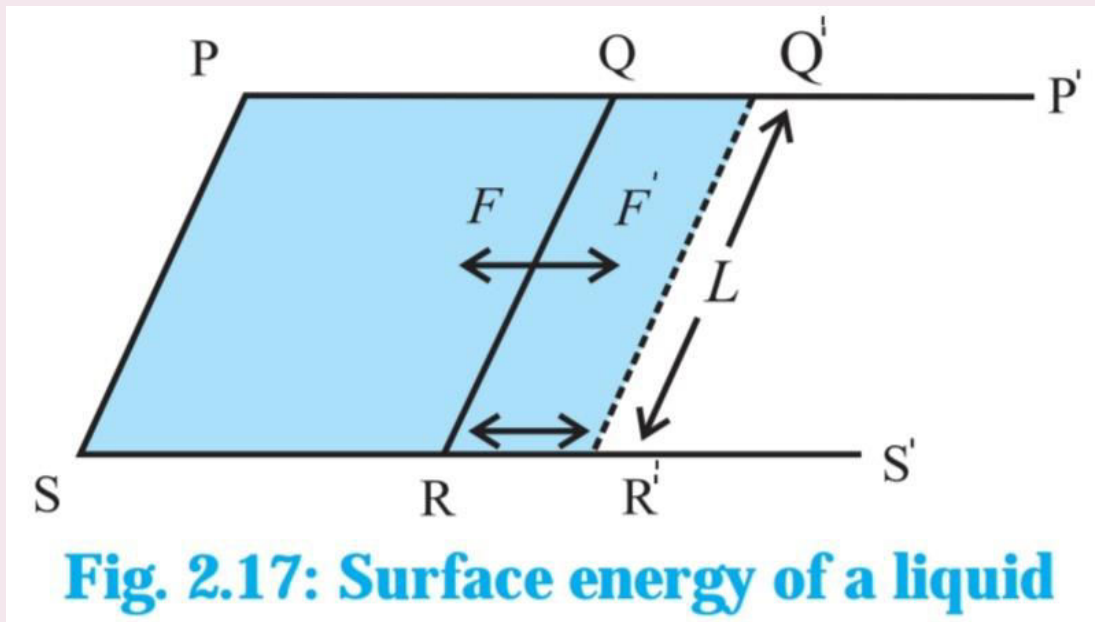


Fig. 2.17: Surface energy of a liquid

⊙ Each arm of the frame experiences an inward force due to the film. Under the action of this force, the movable arm QR moves towards side PS so as to decrease the area of the film.

⊙ If the length of QR is L , then this inward force F acting on it is given by

$$F = (T) \times (2L) \quad \text{--- (1)}$$

⊙ Since the film has two surfaces, the upper surface and the lower surface, the total length over which surface tension acts on QR is $2L$.

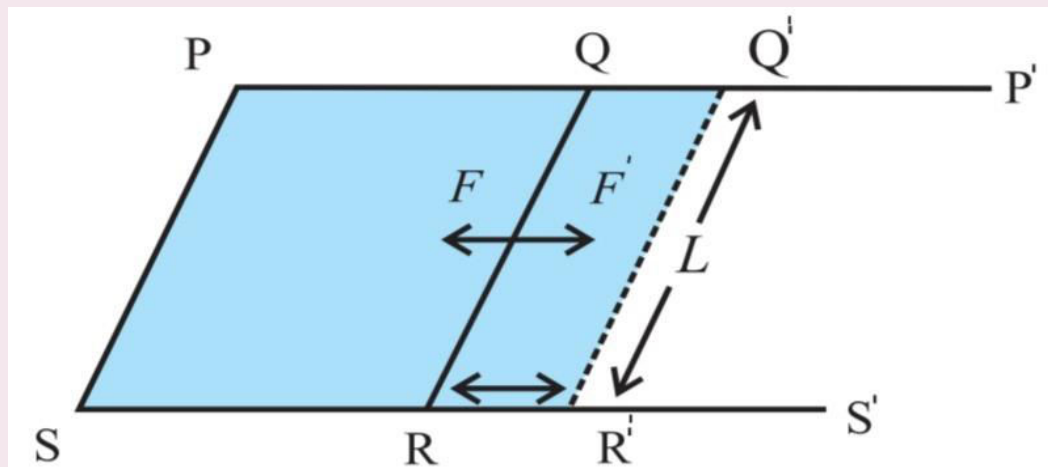


Fig. 2.17: Surface energy of a liquid

- Imagine an external force F' (equal and opposite to F) applied at constant temperature, to the arm QR, so that it pulls the arm away and tries to increase the surface area of the film.
- The arm QR moves to $Q'R'$ through a distance dx . Therefore, the work done against F , the force due to surface tension, is given by

$$dw = F' dx$$

Using Eq. (1),

$$dw = T (2Ldx)$$

- But, $2Ldx = dA$, increase in area of the two surfaces of the film.

Therefore, $dw = T(dA)$.

- This work done in stretching the film is stored in the area dA of the film as its potential energy. *This energy is called surface energy.*

$\therefore$ Surface energy = $T (dA)$ --- (2)

- Thus, *surface tension is also equal to the surface energy per unit area.*

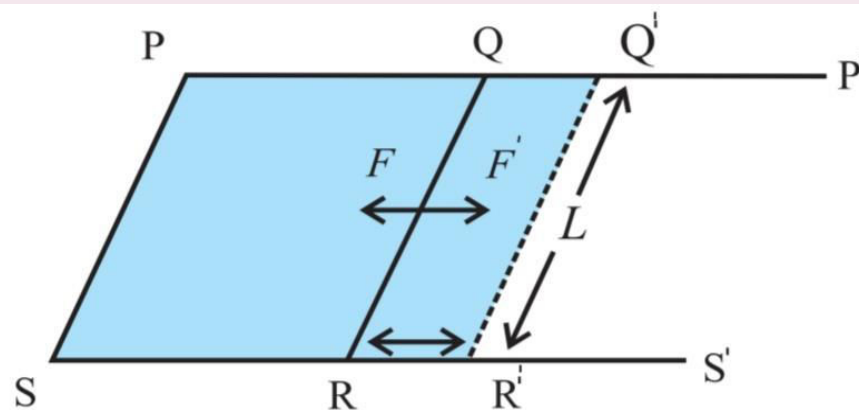


Fig. 2.17: Surface energy of a liquid